BART Model Loss Calculation

The formula for the loss L is:

$$L = -\frac{1}{T}\sum_{t=1}^{T} \ln(P(y_t|y_{<t}, X))$$

where T is the length of the target sequence, and $P(y_t|y_{<t}, X)$ is the model's predicted probability of the correct token y_t at timestep t.

1. Vocabulary and Target Sequence

First, we establish the vocabulary and map each word to an index.

$\mathcal{V} = \{$[start], building, character, a, is, astronomy, science, experience, natural, [end]$\}$

The corresponding indices are:

- `[start]`: 0
- `building`: 1
- `character`: 2
- `a`: 3
- `is`: 4
- `astronomy`: 5
- `science`: 6
- `experience`: 7
- `natural`: 8
- `[end]`: 9

The original (target) sentence is $Y =$ [start] astronomy is a character building experience [end]. The sequence of tokens to be predicted by the decoder has length $T = 7$. The tokens and their indices are:

$$y = (\underbrace{\text{astronomy}}_{y_1,\text{idx}=5}, \underbrace{\text{is}}_{y_2,\text{idx}=4}, \underbrace{\text{a}}_{y_3,\text{idx}=3}, \underbrace{\text{character}}_{y_4,\text{idx}=2}, \underbrace{\text{building}}_{y_5,\text{idx}=1}, \underbrace{\text{experience}}_{y_6,\text{idx}=7}, \underbrace{\text{[end]}}_{y_7,\text{idx}=9})$$

2. Extracting Probabilities

We use the given prediction probability matrix $\hat{Y}$ to find the probability of each correct token in the target sequence. The probability for the target token y_t is found at row $t-1$ and the column corresponding to its index.

$$\hat{Y} = \begin{bmatrix} 0.05 & 0.02 & 0.14 & 0.07 & 0.09 & \mathbf{0.41} & 0.08 & 0.05 & 0.08 & 0.01 \\ 0.10 & 0.04 & 0.06 & 0.16 & \mathbf{0.43} & 0.01 & 0.06 & 0.09 & 0.01 & 0.04 \\ 0.05 & 0.07 & 0.28 & \mathbf{0.29} & 0.03 & 0.08 & 0.04 & 0.08 & 0.04 & 0.04 \\ 0.10 & 0.02 & \mathbf{0.48} & 0.02 & 0.01 & 0.07 & 0.06 & 0.14 & 0.07 & 0.01 \\ 0.08 & \mathbf{0.29} & 0.04 & 0.01 & 0.04 & 0.03 & 0.25 & 0.05 & 0.15 & 0.06 \\ 0.14 & 0.22 & 0.13 & 0.08 & 0.01 & 0.03 & 0.06 & \mathbf{0.23} & 0.08 & 0.03 \\ 0.1 & 0.01 & 0.01 & 0.01 & 0.07 & 0.2 & 0 & 0.25 & 0.05 & \mathbf{0.3} \end{bmatrix}$$

The probabilities for the target tokens are:

- $P(y_1 = \text{"astronomy"}) = \hat{Y}[0, 5] = 0.41$
- $P(y_2 = \text{"is"}) = \hat{Y}[1, 4] = 0.43$
- $P(y_3 = \text{"a"}) = \hat{Y}[2, 3] = 0.29$
- $P(y_4 = \text{"character"}) = \hat{Y}[3, 2] = 0.48$
- $P(y_5 = \text{"building"}) = \hat{Y}[4, 1] = 0.29$
- $P(y_6 = \text{"experience"}) = \hat{Y}[5, 7] = 0.23$
- $P(y_7 = \text{"[end]"}) = \hat{Y}[6, 9] = 0.30$

3. Final Loss Calculation

Now, we compute the negative log-likelihood for each of these probabilities and sum them up.

$$\begin{aligned} \sum_{t=1}^{7} -\ln(P(y_t)) &= -\ln(0.41) - \ln(0.43) - \ln(0.29) - \ln(0.48) - \ln(0.29) - \ln(0.23) - \ln(0.30) \\ &\approx -(-0.8916) - (-0.8440) - (-1.2379) - (-0.7340) - (-1.2379) - (-1.4697) - (-1.2040) \\ &\approx 0.8916 + 0.8440 + 1.2379 + 0.7340 + 1.2379 + 1.4697 + 1.2040 \\ &\approx 7.6191 \end{aligned}$$

Finally, we average this sum by the number of tokens, $T = 7$.

$$L = \frac{7.6191}{7} \approx 1.0884$$